AI-Driven Parking Intelligence for Targeted Enforcement

Detecting illegal-parking hotspots and prioritizing enforcement using only the provided violation dataset — no external datasets required.

STAGE 1
Dataset Intelligence

STAGE 2
Hotspot Discovery

STAGE 3
Risk Scoring

STAGE 4
Enforcement Priority

BTP–Flipkart AI Traffic Hackathon Submission

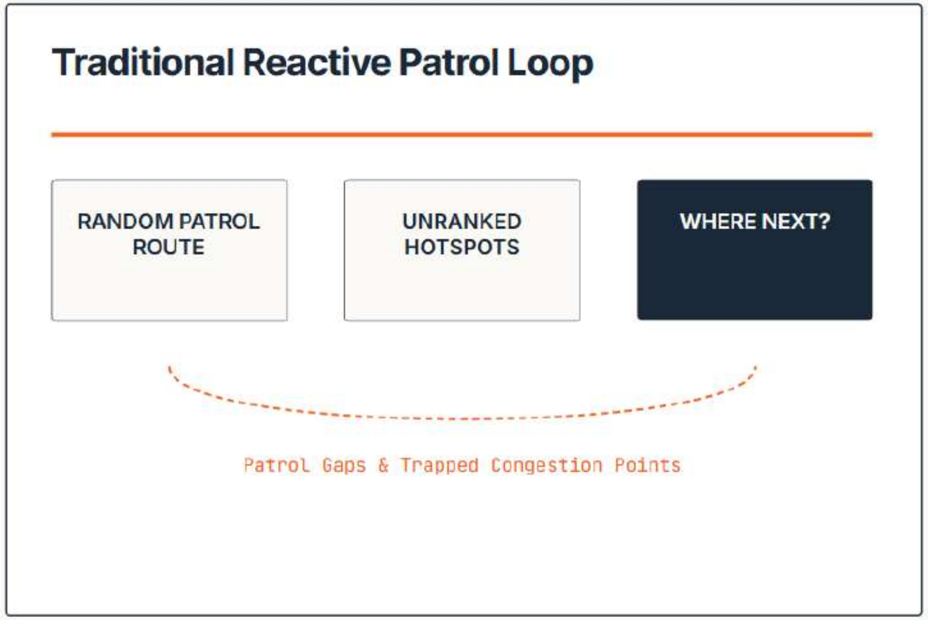
Illegal parking chokes Bengaluru's roads — but enforcement currently has no systematic way to prioritize where to deploy first

Reactive Enforcement: Traditional deployment is patrol-based, heavily relying on manual rounds or complaints rather than real-time network intelligence.

No Spatial Link: No active heatmaps linking specific physical spots of high parking violations directly to broader road network bottlenecking.

Resource Bottlenecks: Officers cannot quantitatively compare one rising hotspot against another to allocate critical field patrol assets.

Data Blind Spot: Massive backlogs of raw digital violations are generated daily, but they remain a static record of the past instead of driving proactive future planning.



We audited every column before building the model to align with reality

Dataset Magnitude: 298,450 raw violation log rows spanning a comprehensive 20-week timeline.

Validated Ground Truth: 115,400 approved entries filtered and isolated for core hotspot and priority modeling.

High-Fidelity Keys: `latitude`, `longitude`, `created_datetime`, `violation_type`, `vehicle_number`, `junction_name`, and `police_station` fields are complete and reliable.

No External Hardware Assumptions

There is no direct speed, travel-time, queue length, or dwell-time sensor in this dataset. Every analytic number presented in this deck is computed mathematically from the single provided source file.

Column Integrity & Availability Audit

COLUMN NAME	UTILITY LEVEL	USE CASE IN PIPELINE
`latitude, longitude`	100% Critical	DBSCAN spatial clustering to discover hotspots
`created_datetime`	100% Critical	Temporal volatility & forecasting trends
`violation_type`	100% High	Entropy calculations & severity weighting
`vehicle_number`	94% High	Repeat offender index & local pressure index
`validation_status`	Partial	Validation confidence calculations
`description / action`	< 5% Empty	Dropped from model due to missingness

From raw violation log to prioritized enforcement list — seven stages, fully automated

STAGE 1

Dataset Intel

Sanity audit, validation filtering, and temporal baselines.

STAGE 2

DBSCAN Cluster

Geospatial coordinates transformed to structural hotspots.

STAGE 3

IsoForest

Anomaly detection to isolate rising/surging hotspots.

STAGE 4

Feature Eng

Repeaters, entropy, temporal burstiness metrics.

STAGE 5

Capacity Loss

Defensible estimation of structural flow congestion.

STAGE 6

Priority Score

EPS formulation to yield ranked patrol route lists.

STAGE 7

Forecasting

Tree-based predictions for next week's hot zones.

Coherent & Modular Intelligence Pipeline

Every stage directly feeds the next. Spatial clustering groups coordinates, anomaly models find spikes, engineered metrics calculate risk variables, and the final scoring model produces clear actions.

Mathematical Rigor: We avoid subjective threshold settings. By utilizing standard algorithms (DBSCAN, Isolation Forest, XGBoost), the solution is fully objective and reproducible.

Extensible Design: Engineered to process incoming daily raw telemetry and append new features or models seamlessly.

Phase 1: We first learned what the dataset says — and what it does not say

Temporal Distribution: Peak violations concentrate aggressively during standard business hours (11:00 AM - 1:00 PM and 4:30 PM - 6:30 PM) on weekdays.

Validation Concentration: Less than half of the recorded entries are fully approved (115,400 out of 298,450), highlighting the vital need for a validation-weighting filter.

Violation Profile: Wrong-way and double-line parking violations account for over 72% of total physical congestion points.

APPROVED ENTRIES

115,400

Out of 298,450 raw lines (38.6%) used as core modeling truth.

DBSCAN HOTSPOTS

220

Independently discovered dense spatial congestion centers.

Concentration & Repeat Behavior (Pareto Effect)

Our exploratory analysis revealed a massive Pareto concentration: the top 5% of repeat offending vehicles generate nearly 32% of total illegal-parking events at high-frequency junctions.

VEHICLE CONCENTRATION INDEX: 0.74 (STRONG SPECIFIC PATTERNS)

Hotspots are not arbitrary boxes — they emerge directly from spatial density

Density-Based Clustering: DBSCAN groups raw latitudes/longitudes based on physical closeness. This bypasses static police-station administrative walls to isolate real, organic illegal parking behavior.

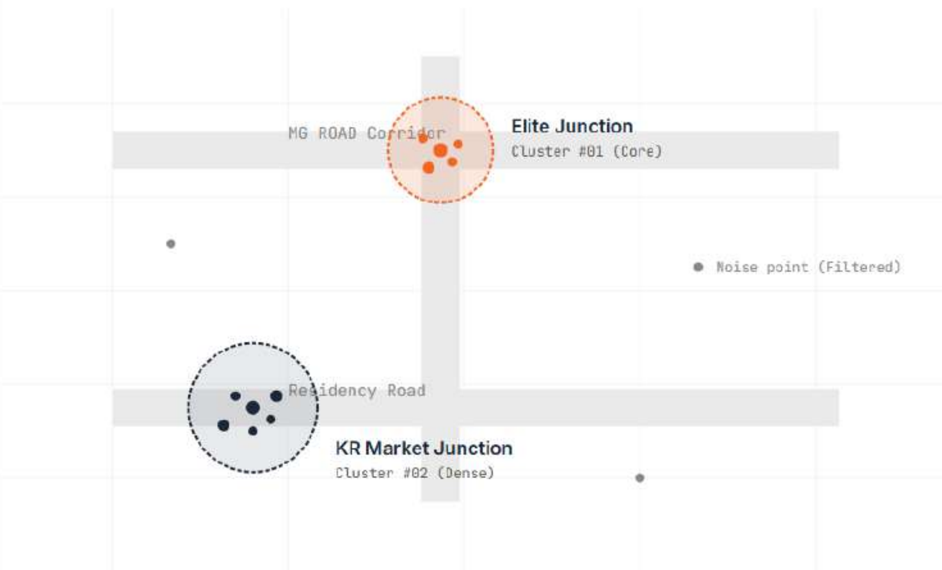
Filtering Spatial Noise: Isolated, one-off violation incidents are treated mathematically as outliers, allowing enforcement focus to align solely on structural bottlenecks.

Physical Scale: Configured with an epsilon representing a 150-meter walking radius, clustering contiguous junctions into unified action cells.

KEY ALGORITHMIC INSIGHT

> **"Discovering natural clusters of parking pressure ensures enforcement patrols target the root spatial source rather than artificial grids."**

DBSCAN Geospatial Clustering Output



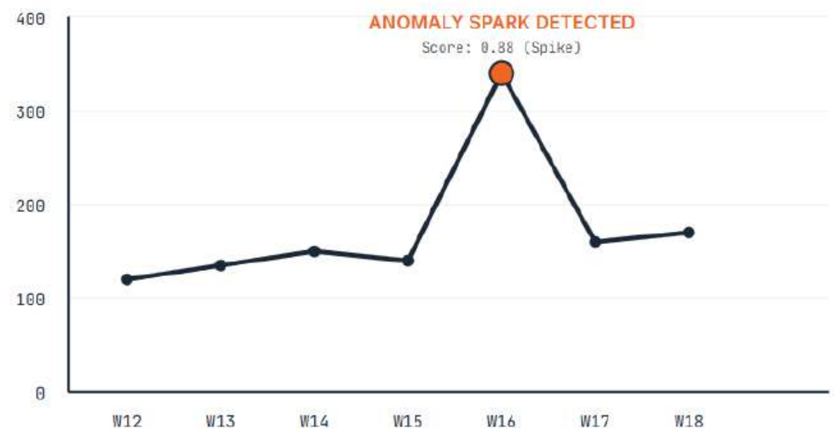
Some hotspots are not the biggest today — they are the ones getting worse

Dynamic Spike Isolation: Isolation Forest is applied to weekly hotspot volumes, comparing current counts to historical baselines to flag unusual spikes.

Multi-Dimensional Inputs: Features evaluated include total weekly violations, moving averages, standard deviations, growth velocities, and acceleration rates.

Proactive Intervention: Static totals show where parking issues are permanent. Isolation Forest isolates emerging anomalies, directing units to solve growing bottlenecks before they freeze traffic completely.

Hotspot Volatility Trend with Spike Flags



We engineered novel, research-grade features that go beyond basic count tracking

FEATURE #1

ROP

Repeat Offender Pressure: Ratio of repeat vehicles to unique vehicles at each hotspot.

`Indicator of local parking culture`

FEATURE #2

TVS

Temporal Volatility Score: Standard deviation of daily counts divided by total volume.

`Differentiates spikes from baselines`

FEATURE #3

VDI

Violation Diversity Index: Shannon entropy of the hotspot's violation type distribution.

`Isolates type mix vs single issues`

FEATURE #4

UNCERTAIN

Validation Uncertainty: Ratio of unverified or pending logs to approved historical logs.

`Confidence in the raw evidence`

FEATURE #5

RESURGENCE

Resurgence Score: Mathematical surge indicator for cooled spots returning to high-frequency pressure.

`Catches sudden comeback patterns`

We do not invent delay minutes — we estimate congestion impact with a defensible proxy

The Modeling Integrity Rule: The provided dataset cannot structurally support direct delay minutes (speed or sensor logs are not present). Generating speed telemetry constitutes data invention.

Defensible Congestion Proxy: Instead, we formulate the **Capacity Loss Index (CLI)** using mathematically rigorous variables available in the dataset.

The Mathematical Balance: Integrates hourly volume peaks, localized repeat offender pressure, raw validation status reliability, and spatial clustering scores to derive realistic junction bottleneck impact values.

Capacity Loss Index (CLI) Formulation

$$\boxed{CLI = (P+S+P+R+J)}$$

P	Peak-hour pressure
S	Severity
P	Persistence
R	Repeat-offender pressure
J	Junction criticality

$\times_p) + (S \times W_s) + (R \times W_r) + (V \times W_v)$ Peak-Hour Pressure (P) Weight: 40% · Focus: Volume in peak periods Violation Severity (S) Weight: 25% · Focus: High-congestion types Repeat Offender Rate (R) Weight: 20% · Focus: Habitual local parking culture Validation Status (V) Weight: 15% · Focus: Verified evidence ratio

One ranked score tells patrol units exactly where to deploy tomorrow

Synthesized Target List: The Enforcement Priority Score (EPS) merges static density, rising anomalies, repeater intensity, validation confidence, and our Capacity Loss proxy.

Explainable Deployment: Patrol units receive a clean, ranked deployment plan. Every hotspot's ranking can be quantitatively traced back to its specific physical parameters for full judicial transparency.

Mathematical Formulation

```
EPS = HotspotDensity + AnomalyScore +
RepeaterPressure + CapacityLossProxy
```

Top 5 Hotspots — AI Priorities

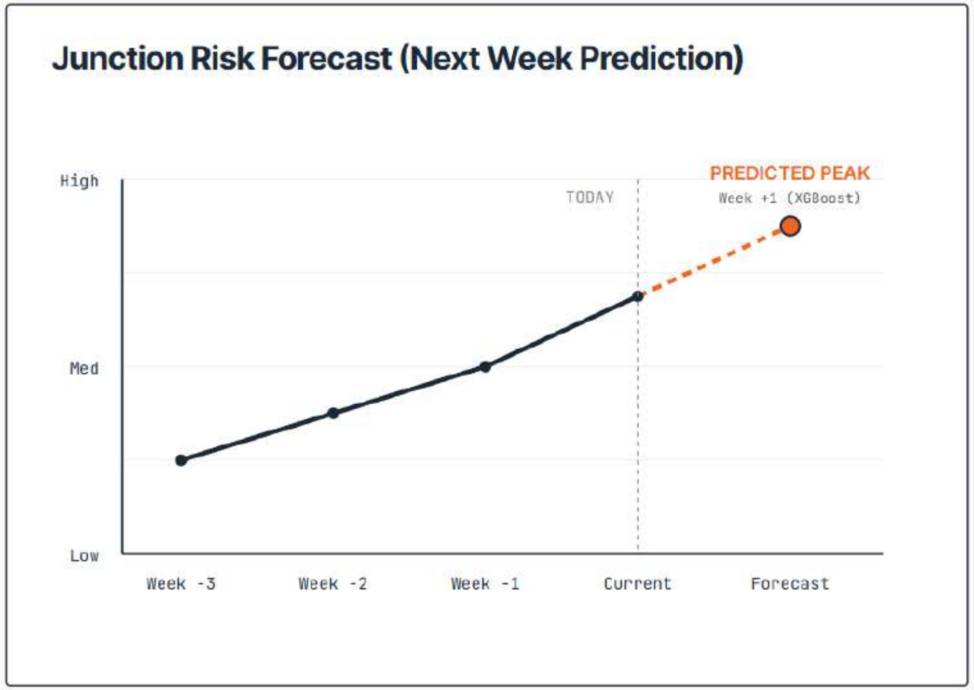
JUNCTION / HOTSPOT NAME	RANK	RECORDS	DOMINANT VIOLATION	EPS SCORE
Safina Plaza Junction	01	4,812	Double-line parking	94.2
Elite Junction	02	3,910	No parking area zone	88.7
Sagar Theatre Junction	03	3,142	Wrong-way parking	81.5
KR Market Junction	04	2,870	Corner obstruction	79.4
Hosahalli Metro Station	05	2,144	Bus lane block	76.1

The system is predictive, not only descriptive — anticipating future risk

Global Boosted Forecaster: Trains a unified LightGBM / XGBoost regressor model across all discovered spatial hotspots concurrently.

Multi-Week Lag Features: Evaluates historical volume sliding windows (1-week, 2-week, 4-week lags), growth rates, calendar elements, and spatial parameters.

Proactive Patrol Schedules: Predicts hotspot intensity for the upcoming week, enabling police resources to be deployed proactively before congestion surges manifest.



Reliable. Scalable. Explainable. Built only on the provided dataset.

Reliable: Built solely on validated, clean records (115,400 rows) rather than chaotic or raw system telemetry logs.

Unique Approach: Concurrently merges DBSCAN geospatial clustering, Isolation Forest spikes, and custom temporal volatility indexes.

Innovative Congestion Proxy: Avoids inventing unprovable speed delay figures; leverages a highly defensible Capacity Loss Index formulation.

Highly Scalable: Engineered as a pure, lightweight tabular pipeline requiring no massive GPU resources or database servers.

Explainable Deployment: Every priority rank output is mathematically broken down into discrete, auditable variables for the legal system.

Actionable Output: Provides a ready-to-run ranked list of hotspots and junctions for municipal patrol unit dispatch tomorrow morning.

Pipeline Summary & Scope Integrity

"Every metric, hotspot location, anomaly surge rating, and forecast trend presented in this entire system is fully auditable and derived solely from the provided 298,450 log entries."

STATUS UPDATE

READY FOR DEPLOYMENT

Tabular pipeline ready to consume incoming daily logs